Chapter Fourteen

TWO BEAM INTERFERENCE BY DIVISION OF WAVEFRONT

The wave nature of light was demonstrated convincingly for the first time in 1801 by Thomas Young by a wonderfully simple experiment…He let a ray of sunlight into a dark room, placed a dark screen in front of it, pierced with two small pinholes, and beyond this, at some distance a white screen. He then saw two darkish lines at both sides of a bright line, which gave him sufficient encouragement to repeat the experiment, this time with spirit flame as light source, with a little salt in it, to produce the bright yellow sodium light. This time he saw a number of dark lines, regularly spaced; the first clear proof that light added to light can produce darkness. This phenomenon is called interference. Thomas Young had expected it because he believed in the wave theory of light.

—Dennis Gabor in his Nobel Lecture, December 11, 1971

Thomas Young had amazing broad interests and talents … From his discoveries in medicine and science, Helmholtz concluded: 'His was one of the most profound minds that the world has ever seen.'

—From the Internet

Learning Objectives

After reading this chapter, the reader should be able to:

LO 1: explain the interference pattern produced on water surface.
LO 2: understand the coherence phenomenon.
LO 3: discuss interference patterns produced by light waves.
LO 4: understand Young's interference pattern and obtain expression for fringe width.
LO 5: discuss intensity distribution in the interference pattern.
LO 6: illustrate Fresnel's two-mirror arrangement and biprism to produce interference pattern.
LO 7: describe interference pattern when the slit is illuminated by white light.
LO 8: discuss displacement of fringes in the interference pattern when a thin sheet is introduced in one of the beams.
LO 9: illustrate Lloyd's mirror arrangement for producing interference patterns.
LO 10: understand the phase change that occurs when a light beam undergoes reflection.

14.1 INTRODUCTION

In the previous chapter, we had considered the superposition of one-dimensional waves propagating on a string and had shown that there is a variation of energy density along the length of the string due to the interference of two waves (see Fig. 13.5). In general, whenever two waves superpose, one obtains an intensity distribution which is known as the interference pattern. In this chapter, we will consider the interference pattern produced by waves emanating from two point sources. It may be mentioned that with sound waves the interference pattern can be observed without much difficulty because the two interfering waves maintain a constant phase relationship; this is also the case for microwaves. However, for light waves, due to the very process of emission, one cannot observe interference between the waves from two independent sources,* although the interference does take place (see Sec. 14.4). Thus, one tries to derive interfering

* It is difficult to observe the interference pattern even with two laser beams unless they are phase locked.

waves from a single wave so that the phase relationship is maintained. The methods to achieve this can be classified under two broad categories. Under the first category, in a typical arrangement, a beam is allowed to fall on two closely spaced holes and the two beams emanating from the holes interfere. This method is known as **division of wavefront** and will be discussed in detail in this chapter. In the other method, known as **division of amplitude**, a beam is divided at two or more reflecting surfaces and the reflected beams interfere. This will be discussed in the next chapter. We must, however, emphasize that the present and the following chapters are based on one underlying principle, namely the superposition principle.

It is also possible to observe interference using multiple beams; this is known as **multiple beam interferometry** and will be discussed in Chapter 16. It will be shown that multiple beam interferometry offers some unique advantages over two beam interferometry.

14.2 INTERFERENCE PATTERN PRODUCED ON THE SURFACE OF WATER LO 1

We consider surface waves emanating from two point sources in a water tank. We may have, for example, two sharp needles vibrating up and down at the points S_1 and S_2 (see Fig. 14.1). Although water waves are not strictly transverse, we will, for the sake of simplicity, assume water waves to produce displacements which are transverse to the direction of propagation.

If there was only one needle (say at S_1) vibrating with a certain frequency v then circular ripples would have spread out from the point S_1. The wavelength would have been v/v and the crests and troughs would have moved outwards (see Fig. 16 in the prelim pages). Similarly for the vibrating needle at S_2. However, if both needles are vibrating, then waves emanating from S_1 will interfere with the waves emanating from S_2. We assume that the needle at S_2 vibrates in phase with the needle at S_1, i.e., S_1 and S_2 go up simultaneously, they also reach the lowest position at the same time. Thus, if at a certain instant, the disturbance emanating from the source S_1 produced a crest at a distance ρ from S_1 then the disturbance from S_2 would also produce a crest at a distance ρ from S_2, etc. This is explicitly shown in Fig. 14.1, where the solid curves represent (at a particular instant) the positions of the crests due to disturbances emanating from S_1 and S_2. Similarly, the dashed curves represent (at the same instant) the positions of the troughs. Notice that at all points on the perpendicular bisector OZ the disturbances reaching

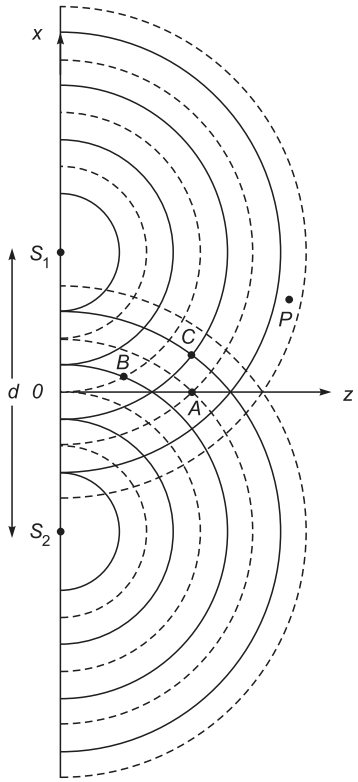


Fig. 14.1 Waves emanating from two pointsources S_1 and S_2 vibrating in phase. The solid and the dashed curves represent the positions of the crests and troughs, respectively.

from S_1 and from S_2 will always be in phase. Consequently, at an arbitrary point A (on the perpendicular bisector) we may write the resultant disturbance as

$$y = y_1 + y_2 = 2a \cos \omega t \qquad \textbf{(14.1)}$$

where y_1 (= $a \cos \omega t$) and y_2 (= $a \cos \omega t$) represent the displacements at the point A due to S_1 and S_2, respectively. We see that the amplitude at A is twice the amplitude produced by each one of the source. It should be noted that at $t = \frac{T}{4}\left(=\frac{1}{4v}=\frac{\pi}{2\omega}\right)$ the displacements produced at the point A by each of the source would be zero and the resultant will also be zero. This is also obvious from Eq. (14.1).

Next, let us consider a point B such that

$$S_2B - S_1B = \lambda/2 \qquad \textbf{(14.2)}$$

At such a point the disturbance reaching from the source S_1 will always be out of phase with the disturbance reaching from S_2. This follows from the fact that the disturbance reaching the point B from the source S_2 must have started half a period (= $T/2$) earlier than the disturbance reaching B

from S_1. Consequently, if the displacement at B due to S_1 is given by

$$y_1 = a \cos \omega t$$

then the displacement at B due to S_2 would be given by

$$y_2 = a \cos (\omega t - \pi) = -a \cos \omega t$$

and the resultant $y = y_1 + y_2$ is zero at all times. Such a point corresponds to destructive interference and is known as a node and corresponds to minimum intensity. It may be pointed out that the amplitudes of the two vibrations reaching the point B will not really be equal as it is at different distances from S_1 and S_2. However, if the distances involved are large (in comparison to the wavelength), the two amplitudes will be very nearly equal and the resultant intensity will be very nearly zero.

In a similar manner, we may consider a point C such that

$$S_2C - S_1C = \lambda$$

where the phase of the vibrations (reaching from S_1 and S_2) are exactly the same as at the point A. Consequently, we will again have constructive interference. In general, if a point P is such that

$$S_2P \sim S_1P = n\lambda \text{ (maxima)} \qquad \textbf{(14.3)}$$

$n = 0, 1, 2, \ldots$, then the disturbances reaching the point P from the two sources will be in phase, the interference will be constructive and the intensity will be maximum. On the other hand, if the point P is such that

$$S_2P \sim S_1P = \left(n + \frac{1}{2}\right)\lambda \text{ (minima)} \qquad \textbf{(14.4)}$$

then the disturbances reaching the point P from the two sources will be out of phase, the interference will be destructive and the intensity will be minimum. The actual interference pattern produced from two point sources vibrating in phase in a ripple tank is shown in Fig. 14.2.

Example 14.1 The intensity at the point which neither satisfies Eq. (14.3) nor Eq. (14.4) will neither be a maximum nor zero. Consider a point P such that $S_2P - S_1P = \lambda/3$. Find the ratio of the intensity at the point P to that at a maximum.

Solution: If the disturbance reaching the point P from S_1 is given by

$$y_1 = a \cos \omega t$$

then the disturbance from S_2 would be given by

$$y_2 = a \cos \left(\omega t - \frac{2\pi}{3}\right)$$

because a path difference of $\lambda/3$ corresponds to a phase difference of $\frac{2\pi}{3}$. Thus the resultant displacement would be

$$y = y_1 + y_2$$

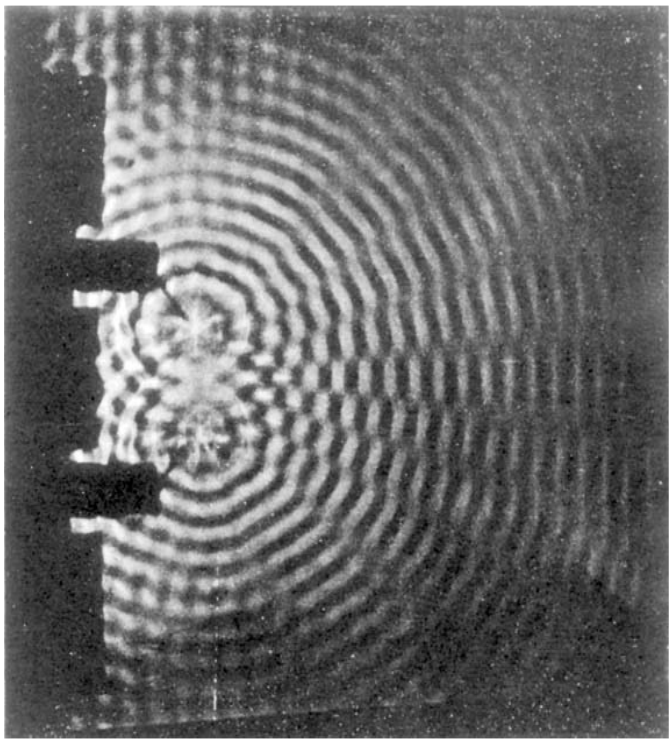


Fig. 14.2 The actual interference pattern produced from two point sources vibrating in phase in a ripple tank (After Ref. 14.9, used with permission).

$$= a\left[\cos \omega t + \cos\left(\omega t - \frac{2\pi}{3}\right)\right]$$

$$= 2a \cos\left(\omega t - \frac{\pi}{3}\right)\cos\frac{\pi}{3}$$

$$= a \cos\left(\omega t - \frac{\pi}{3}\right)$$

The intensity is therefore 1/4th of the intensity at the maxima. In a similar manner, one can calculate the intensity at any other point.

Example 14.2 The locus of points which correspond to minima are known as nodal lines. Show that the equation of a nodal line is a hyperbola. Also obtain the locus of points which correspond to maxima.

Solution: For the sake of generality, we find the locus of the point P which satisfies the following equation:

$$S_2P - S_1P = \Delta \tag{14.5}$$

Thus, if $\Delta = n\lambda$, we have a maximum and if $\Delta = \left(n + \frac{1}{2}\right)\lambda$ we have a minimum. We choose the midpoint of S_1S_2 as the origin, with the x-axis along S_1S_2 and the z-axis perpendicular to it (see Figs. 14.1 and 14.3). We will find the locus of the point P on the x-z plane. If the distance between S_1 and S_2 is d, then the coordinates of the points S_1 and S_2 are $\left(+\frac{d}{2}, 0\right)$ and $\left(-\frac{d}{2}, 0\right)$, respectively. Let the coordinates of the point P [see Fig. 14.3] be (x, z). Then,

$$S_1P = \left[\left(x - \frac{d}{2}\right)^2 + z^2\right]^{1/2}$$

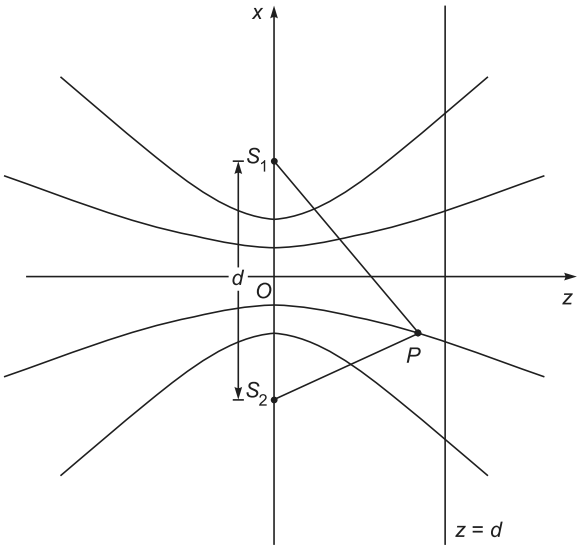


Fig. 14.3 The nodal curves.

and

$$S_2P = \left[\left(x + \frac{d}{2}\right)^2 + z^2\right]^{1/2}$$

Therefore,

$$S_2P - S_1P = \left[\left(x + \frac{d}{2}\right)^2 + z^2\right]^{1/2} - \left[\left(x - \frac{d}{2}\right)^2 + z^2\right]^{1/2} = \Delta$$

or

$$\left(x + \frac{d}{2}\right)^2 + z^2 = \left(x - \frac{d}{2}\right)^2 + z^2 + \Delta^2 + 2\Delta\left[\left(x - \frac{d}{2}\right)^2 + z^2\right]^{1/2}$$

or

$$2xd - \Delta^2 = 2\Delta\left[\left(x - \frac{d}{2}\right)^2 + z^2\right]^{1/2}$$

On squaring, and simplifying we obtain

$$\frac{x^2}{\frac{1}{4}\Delta^2} - \frac{z^2}{\frac{1}{4}(d^2 - \Delta^2)} = 1 \tag{14.6}$$

which is the equation of a hyperbola. When $\Delta = \left(n + \frac{1}{2}\right)\lambda$ the curve correspond to minima and when $\Delta = n\lambda$ the curve correspond to maxima. For large value of x and z the curve asymptotically tend to the straight lines

$$z = \pm\left(\frac{d^2 - \Delta^2}{\Delta^2}\right)^{1/2} x \tag{14.7}$$

It may be pointed out that $S_2P \sim S_1P$ equals d only on the x-axis and that there is no point P for which $S_2P \sim S_1P > d$. Now, it appears from Eq. (14.6) that when $\Delta > d$, the resulting equation is an ellipse which we know is impossible. The fallacy is a result of the fact that because of a few squaring operations, Eq. (14.6) also represents the locus of all those points for which $S_2P + S_1P = \Delta$ and obviously in this case Δ can exceed d.

Example 14.3 Consider a line parallel to the x-axis at a distance D from the origin (see Fig. 14.3). Show that for path differences small compared to d, the points on the line where minimum intensity occurs are equally spaced.

Solution: The equation of the line would be

$$z = D \tag{14.8}$$

Further at large distances from the origin the equation of the nodal lines would be